

WASP-36b

R-0843

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_230217 · created 2026-07-28T19:28:41+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459992.8302 +/- 0.0051 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.135 +/- 0.031
Transit depth (Rp/Rs)^2	1.82 +/- 0.84 [%]
Orbital Inclination (inc)	81.77 +/- 2.43
Airmass coefficient 1 (a1)	0.962 +/- 0.016
Airmass coefficient 2 (a2)	0.0263 +/- 0.0072
Scatter in the residuals of the lightcurve fit is	1.69 %
Best Comparison Star	#3 - [456, 167]
Optimal Method	PSF photometry
Transit Duration (day)	0.058 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.135 ± 0.031 vs 0.1368 (deviation 0.06σ)
Duration fit vs published	0.058 d vs 0.0758 d (deviation 0.66σ)
Mid-time O–C	-12.3 min
Depth SNR	4.4
Residual scatter	1.69 %

Light curve

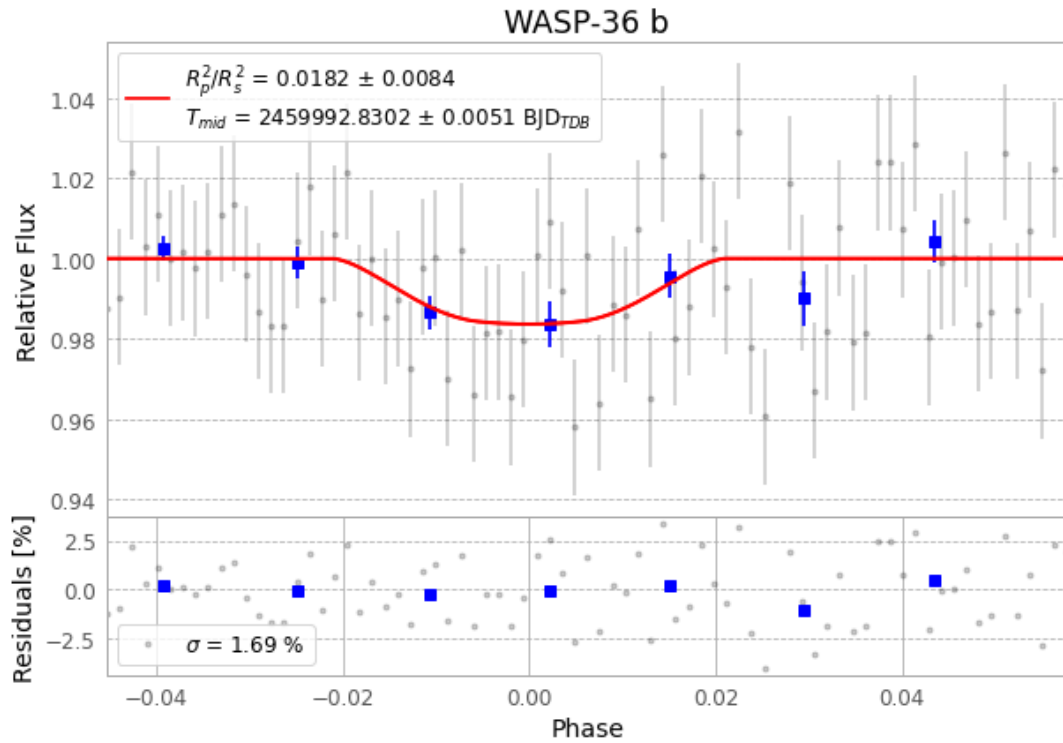


Figure 1 — FinalLightCurve_WASP-36 b_2023-02-16.png (SHA-256 2550da0af8087b65...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [388, 199], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 254a9bebbc1d563e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

77 FITS frames (51 MB), WASP-36230217060615.FITS → WASP-36230217095415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-230217.log (a1dd647211ca...), FinalLightCurve_WASP-36 b_2023-02-16.png (2550da0af808...), FinalLightCurve_WASP-36 b_2023-02-16.csv (1dd1d7aaa681...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 19:28Z.